

Limits Mark Scheme A

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Award **M** marks for a correct method, **A** marks for the accurate result that follows. A 26 marks · 30 min correct answer with no working scores **0**.

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
8	$\frac{1}{8}$	$\frac{9}{2}$	-12	$\frac{3}{4}$	2	1	$\frac{1}{2}$

1

$$\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$$

= 8

$$= \lim_{x \rightarrow 4} \frac{(x - 4)(x + 4)}{x - 4} \quad \text{M1}$$

Substitution gives 0/0, so factorise the difference of two squares.

$$= \lim_{x \rightarrow 4} (x + 4), \quad x \neq 4 \quad \text{A1}$$

$$= 4 + 4 = \mathbf{8} \quad \text{A1}$$

2

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+16} - 4}{x}$$

= $\frac{1}{8}$

$$= \lim_{x \rightarrow 0} \frac{(\sqrt{x+16} - 4)(\sqrt{x+16} + 4)}{x(\sqrt{x+16} + 4)} \quad \text{M1}$$

Multiply above and below by the conjugate of the numerator.

$$= \lim_{x \rightarrow 0} \frac{(x + 16) - 16}{x(\sqrt{x+16} + 4)} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+16} + 4} \quad \text{A1}$$

$$= \frac{1}{4 + 4} = \frac{\mathbf{1}}{\mathbf{8}} \quad \text{A1}$$

$$\boxed{3} \quad \lim_{x \rightarrow 3} \frac{x^3 - 27}{x^2 - 9}$$

$$= \frac{9}{2}$$

$$= \lim_{x \rightarrow 3} \frac{(x-3)(x^2 + 3x + 9)}{(x-3)(x+3)}$$

M1

Difference of two cubes above, difference of two squares below.

$$= \lim_{x \rightarrow 3} \frac{x^2 + 3x + 9}{x + 3}$$

A1

$$= \frac{9 + 9 + 9}{6} = \frac{9}{2}$$

A1

$$\boxed{4} \quad \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 5x + 6}$$

$$= -12$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)}{(x-2)(x-3)}$$

M1

$$= \lim_{x \rightarrow 2} \frac{x^2 + 2x + 4}{x - 3}$$

A1

$$= \frac{4 + 4 + 4}{2 - 3} = \frac{12}{-1} = -12$$

A1

WATCH FOR the denominator $2 - 3 = -1$. Answers of $+12$ lose the final A mark.

$$\boxed{5} \quad \lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 2}{4x^2 + 7x - 1}$$

$$= \frac{3}{4}$$

$$= \lim_{x \rightarrow \infty} \frac{3 - 5/x + 2/x^2}{4 + 7/x - 1/x^2}$$

M1

Divide every term above and below by x^2 , the highest power present.

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$$

A1

$$= \frac{3 - 0 + 0}{4 + 0 - 0} = \frac{3}{4}$$

A1

6

$$\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + 1}}{x + 3}$$

= 2

$$\sqrt{4x^2 + 1} = x \sqrt{4 + 1/x^2} \text{ for } x > 0$$

M1

Take x^2 out of the root: $\sqrt{\frac{\quad}{x^2}} = \frac{\quad}{|x|} = \frac{\quad}{x}$ as $x \rightarrow \infty$.

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{4 + 1/x^2}}{1 + 3/x}$$

A1

$$= \frac{\sqrt{4}}{1} = 2$$

A1

WATCH FOR as $x \rightarrow -\infty$ the same limit is -2 , because $\sqrt{\frac{\quad}{x^2}} = -\frac{\quad}{x}$ there. Worth a minute at the board.

7

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$$

= 1

$$= \lim_{x \rightarrow 0} \frac{(\sqrt{1+x} - \sqrt{1-x})(\sqrt{1+x} + \sqrt{1-x})}{x(\sqrt{1+x} + \sqrt{1-x})}$$

M1

Multiply above and below by the conjugate.

$$= \lim_{x \rightarrow 0} \frac{(1+x) - (1-x)}{x(\sqrt{1+x} + \sqrt{1-x})} = \lim_{x \rightarrow 0} \frac{2x}{x(\sqrt{1+x} + \sqrt{1-x})}$$

A1

$$= \lim_{x \rightarrow 0} \frac{2}{\sqrt{1+x} + \sqrt{1-x}} \quad (\text{cancel } x \neq 0)$$

A1

$$= \frac{2}{1+1} = 1$$

A1

8

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$$

$$= \frac{1}{2}$$

$$= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + x} - x)(\sqrt{x^2 + x} + x)}{\sqrt{x^2 + x} + x}$$

M1

The form is $\infty - \infty$, so rationalise: multiply and divide by the conjugate.

$$= \lim_{x \rightarrow \infty} \frac{(x^2 + x) - x^2}{\sqrt{x^2 + x} + x} = \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + x} + x}$$

A1

$$= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + 1/x} + 1} \quad (\text{divide by } x > 0)$$

A1

$$= \frac{1}{1 + 1} = \frac{1}{2}$$

A1

WATCH FOR students who write $\infty - \infty = 0$. State the indeterminate form before rationalising.